

PROJECT DESIGN PHASE

Metro Ticket Generating System in ServiceNow

Problem–Solution Fit • Proposed Solution • Solution Architecture

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Project Name	Metro Ticket Generating System in ServiceNow

Overview

This phase turns the requirements gathered during ideation into a concrete ServiceNow design. It shows how each passenger pain point is answered by a platform capability, summarizes the proposed solution, and describes the layers and data flow that make up the Metro Ticket Generating System.

1. Problem – Solution Fit

Each recurring passenger problem is paired with the need behind it and the ServiceNow capability chosen to meet it.

Customer Problem	Observed Need	Proposed Capability	Fit / Benefit
Ticket booking is slow and manual	Quick, hassle-free digital booking	“Book A Metro Ticket” Service Catalog item	Cuts booking time and effort
Fares are hard to work out for each route	Correct fare shown before confirming	Catalog Client Script with a station-to-station fare matrix	Fare is calculated automatically
Passengers prefer different ways to pay	A choice of payment methods	UPI / Card / Others, with a UI Policy for extra details	Suits a wide range of users
A ticket must be shown at the gate	A ticket that is easy to display	Metro QR Widget on the Service Portal	QR ticket available digitally
Booking records need to stay traceable	A record kept automatically for every booking	Native ServiceNow Requested Item (RITM)	No extra database or manual record-keeping
Request numbers can be hard to track	Simple, familiar tracking	Standard RITM number generated by ServiceNow	Easy traceability for staff

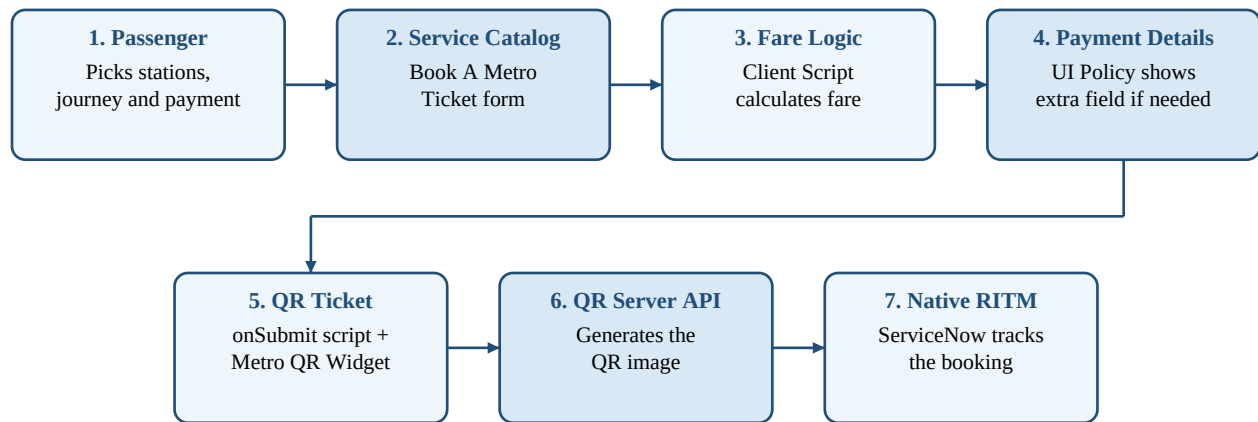
2. Proposed Solution

The table below summarizes the solution against the standard project-design parameters.

S.No.	Parameter	Description
1	Problem Statement	Metro passengers need a simple, quick, and dependable way to book tickets digitally, with the fare worked out correctly and a ticket they can show easily at the gate.
2	Idea / Solution Description	Build a Metro Ticket booking flow on ServiceNow. Passengers open the Service Portal (/sp), search for Book A Metro Ticket , and choose their source and destination stations, journey type, and number of passengers. A Catalog Client Script calculates the fare on the form, a UI Policy shows the extra payment field only when needed, and an onSubmit Client Script generates a QR code shown through the Metro QR Widget. The booking itself is tracked as a standard ServiceNow Requested Item (RITM); no separate database table or Flow Designer automation is used.
3	Novelty / Uniqueness	Brings station-based fare automation, conditional payment details, and QR ticket display together in a single Service Catalog workflow, using only standard ServiceNow features (Service Catalog, Client Scripts, UI Policy, and a Service Portal widget) plus a QR image URL, with no external ticketing product or extra data store.
4	Social Impact / Customer Satisfaction	Shorter waits at counters, less manual work for passengers and staff, support for cashless payment, and less paper because tickets are digital.
5	Business Model (Revenue Model)	Primarily a service-automation solution. In a live metro setting it would support fare collection and smoother station operations. The project brief does not define a specific revenue model.
6	Scalability of the Solution	New stations, fare rules, payment integrations, reports, and real gate-level ticket validation can be added on top of the current workflow without redesigning it.

3. Solution Architecture

The design is organized into layers, from the passenger at the top to the data and security layers underneath. The diagram shows how a booking moves through them.



Layers and Responsibilities

Layer	Component	Responsibility
User Layer	Passengers (primary); Station Managers, Metro Operations Team, IT Admins (supporting)	Passengers choose route, journey, passenger, and payment options; staff oversee operations and configuration.
Experience Layer	Service Portal / Service Catalog	Presents the Book A Metro Ticket form and the booking confirmation.
Logic Layer	Catalog Client Script (onChange)	Validates the chosen route and calculates the fare.
Logic Layer	Catalog UI Policy	Shows or hides the Enter Payment Mode field.
Digital Ticket Layer	Catalog Client Script (onSubmit) + Metro QR Widget + QR Server API	Generates the QR image on submission and displays it through the widget.
Data Layer	Metro Station's Details (u_metro_station_details)	Holds the station master list.
Request Tracking Layer	Native ServiceNow Requested Item (RITM)	Stores the booking as a standard RITM, with journey, fare, payment, and QR details as variables.
Security Layer	ServiceNow ACLs / Roles	Default table ACLs keep the custom station table out of direct reach of portal users.

Key Design Decisions

- **Station reference data:** the Metro Station's Details table (single field, Station Name — *u_station_name*) supplies the choices for Starting From and Going To.
- **Fare on the form:** the fare is calculated on the catalog form from a configured station-to-station fare matrix, so the passenger sees the amount before submitting.

- **No separate database:** booking details are kept as variables on the catalog request itself; there is no custom Metro Database table and no Flow Designer flow moving data into one.
- **Tracking via native RITM:** each submitted booking is tracked using ServiceNow's standard Requested Item (RITM), with journey, fare, payment, and QR details captured as variables on it.
- **Client-side automation:** fare calculation runs on an onChange Client Script, and QR generation runs on an onSubmit Client Script, so no server-side flow is required.
- **QR presentation:** the Metro QR Widget (ID *metro_qr_widget*) receives a QR image URL from the QR Server API and shows it on the portal.
- **Access:** passengers book through the catalog only; the default ACLs created with the station table are used, and stricter role-based rules are a future enhancement.